SULIT

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION
SEMESTER I 2023/2024

CODE OF SUBJECT	**: SECJ3323**
NAME OF SUBJECT	**: SOFTWARE ARCHITECTURE AND DESIGN**
YEAR / COURSE	**: 1/2/3/4SECJ**
DURATION	**: 3 Hours**
DATE	**: 5th February 2024**
VENUE	**:**

THIS EXAMINATION BOOK CONSISTS OF TWO (2) PARTS:

- **PART A**: 4 STRUCTURED QUESTIONS (40 MARKS)
  Answer all the questions

- **PART B**: 2 PROBLEM SOLVING QUESTION (60 MARKS)
  Answer all the questions

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

Name	
I/C number	
Year/Course	
Section	
Lecturer	

This Examination book consist of 9 printed pages excluding this page

PART A: STRUCTURED QUESTION [40 Marks]

Question 1 **[10 marks]**

a. The following is the MediNET System Description document billing function description.

Billing

A bill is issued to each patient after the consultation is completed. Each bill is given a unique bill number as there will not be the same bill number for different bill. The information about clinic such as clinic name and ID are included in a bill. Other information such as user ID of the staff who creates the bill as well as the issue date and time are also included in the bill. The prescriptions supplied by a doctor are used to create bill items for the bill. Each bill contains at least one bill item. The quantity of bill item is supplied to calculate its cost; the amount of the bill is calculated as the sum of all bill items. The bill item has an item code which refers to a specific transaction item. Transaction items are grouped under several transaction types. Among the examples of transaction types are consultation, medication, ward charges and X-Ray. These types are not fixed and may change over time. Each transaction item has an item code to retrieve details on a particular transaction item. The details include measure unit, service tax, and default amount.

Figure 1: MediNET System Description document billing function description

Based on the MediNET System Description document billing function description, write a Programming Design Language that represent the operations/workflow shown in Figure 2.

[5 marks]

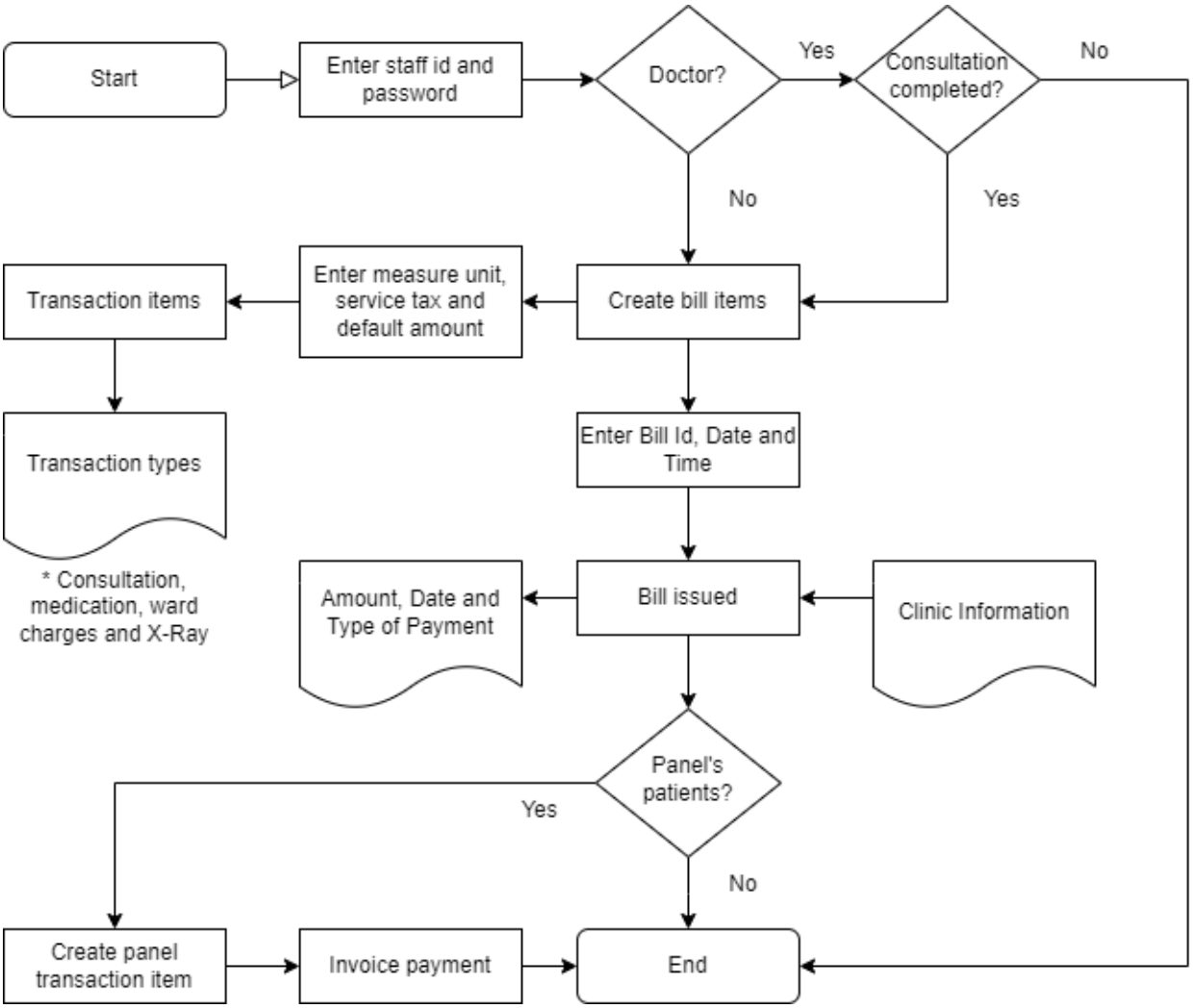


Figure 2: Workflow to Generate a Billing

b. The cyclomatic complexity computation allows designers to measure the complexity of flow-based operational designs by determining the complexity of the decision structure of operations instead of lines of code. Based on Figure 3, convert the codes to draw the program flow Graph and determine the cyclomatic complexity using any approach.

```
public static int sampleFunction(int x, int y) {
    int result = x + y;

    if (result > 0) {
        System.out.println("Positive result");
    } else if (result == 0) {
        System.out.println("Zero result");
    } else {
        System.out.println("Negative result");
    }

    for (int i = 0; i < result; i++) {
        System.out.println("Iteration: " + i);
    }

    return result;
}
```

Figure 3: Sample code

[5 marks]

Question 2 **[10 marks]**

a) Identify **THREE(3)** Decision Tables.Then, sketch a suitable Decision Table based on the Hotel Ayu Villa System Description document on booking business process below.

The Business Processes

In Hotel Ayu Villa, there are four main business processes: booking management, staff scheduling shifts, hotel service and maintenance and customer service. Booking management can be handle by hotel staff to receive any booking made by customer.

Customer need to create an account and login to the system to make any booking. Staff can view, update status, create and delete booking. Staff scheduling shifts can only be updated by staff admin of the hotel. The scheduling needs to consider the occupancy levels and events expected in the hotel. Other requirements that need to be considered are the number of full and part time staff on the payroll and provide compensatory off and holidays. Hotel Ayu Villa need to follow these requirements in order to achieve a good staff scheduling management.

Hotel service and maintenance include hotel environment and hotel room. Maintenance staff will update status of the report made by customer or other staff if the issue has been fixed. As for the customer service, Hotel Ayu Villa gives out promotions and vouchers based on customers booking data to increase hotel sales and attraction.

Figure 4: Hotel Ayu Villa System booking business processes descriptions

[10 marks]

Question 3 **[10 marks]**

a) Describe the term "Work Breakdown Structure" (WBS) in project management and identify **TWO(2)** of its primary purposes.

[4 marks]

b) Based on the Hotel Ayu Villa case study in Figure 4, identify the components and sketch a WBS.

[6 marks]

Question 4 **[10 marks]**

a. Earned Value Management (EVM) is a project management technique to determine the progress of tasks based on the value of the work currently completed versus the work that was expected to be completed at that particular time.

Let's consider a software development project where a team is working on creating a new feature for a software application. The project was estimated to take 10 weeks to complete, with a total budget of $100,000. After 5 weeks, you want to assess the project's progress using EVM.

Situation:

Project Duration Estimate: 10 weeks

Total Budget: $100,000

After 5 weeks:

Planned Value (PV): The planned value of the work that was expected to be completed by this time is estimated to be $50,000

Earned Value (EV): The actual value of the work completed by this time is $45,000.

Actual Cost (AC): The actual cost incurred so far in the project is $55,000.

Figure 5: Project progress case study

i. Based on Figure 5, calculate EVM value.
Note: Formula for EVEM calculation can be referred in Appendix A

[6 marks]

b) Leadership plays a key role in the success or failure of complex projects. There are **FOUR** (4) keys in leadership skills. Based in your own experience, explain how do you apply **TWO** (2) keys in leadership skills during your project in this course.

[4 marks]

PART B: PROBLEM SOLVING **[60 Marks]**

Part B consists of **TWO (2)** essay questions. Answer **all the** questions.

Question 1 **[35 marks]**

Case Study: Weather Monitoring System

Background:
As a software engineer, you are tasked with designing the development of a Weather Monitoring System. The system needs to gather real-time weather data from various sensors and display it in a user-friendly interface. Additionally, the system should notify users whenever there are significant weather changes.The system should be extensible, allowing for the easy addition of new weather sensors in the future. Lastly, the users should be able to customize the way they receive weather alerts, such as through email, SMS, or an in-app notification.

Figure 6: Weather Monitoring System Descriptions

a) Based on Figure 6, imagine you have to integrate a cutting-edge weather sensor into your existing system. This sensor uses a JSON-based data format and communicates over WebSockets, while your current system relies on XML format and RESTful APIs. Consider the challenges posed by the differences in data formats (JSON vs. XML) and communication protocols (WebSockets vs. RESTful APIs). Choose and design pattern and draw a class diagram that not only accommodates the new sensor but also allows for future sensor additions with varying formats and protocols.

[15 marks]

b) Based on Q1a, discuss how the chosen detailed design pattern ensuring that the new sensor can be incorporated without affecting the existing components of the system.

[5 marks]

c) Draw a new class diagram that accommodate new notification channels without requiring significant changes to the existing system architecture.

[15 marks]

Question 2 **[25 marks]**

Case Study: Vehicle Manufacturing System

As a software engineer, you are tasked with designing the development of a modular and extensible Vehicle Manufacturing System. The system should be capable of producing various types of vehicles, such as cars and motorcycles, each with its own set of components. Additionally, the system should support different manufacturing processes for different vehicle types. The system should be able to create families of related objects, like engines and chassis.The design should allow for easy addition of new vehicle types and manufacturing processes in the future.

Figure 7: Vehicle Manufacturing System Descriptions

a) Imagine you are tasked with designing a Vehicle Manufacturing System that needs to support different vehicle types, each requiring unique components like engines and chassis. Draw a class diagram that shows the families of related objects without explicitly specifying their concrete classes. Design the solution that accommodates concrete classes for cars, motorcycles, engines, and chassis.

[10 marks]

b) Write an excerpt code to apply the class diagram drawn based on Question 2a.

[15 marks]

- END QUESTION -

APPENDIX A

Table 1:Earned Value Management Calculation Table

Parameter	Equation
EV	EV = (% work completed) * planned cost for the work
CV	CV = EV - AC
SV	SV = EV - PV
CPI	CPI = EV / AC
SPI	SPI = EV / PV
ETC	ETC = (BAC – EV) / CPI
EAC	EAC = ETC + AC